	rate	
7a	A photon is a quantum of electromagnetic radiation. The energy (of a photon) is dependent only on the frequency of radiation / Each photon has an energy given by the product of Planck constant and the frequency of the radiation / Each photon carries an energy = *hf*	B1 B1
7bi	Light is re-emitted in all directions / only part of the re-emitted light is in the direction of the (original) beam.	B1
7bii	Energy of photon released from the transition (largest energy difference so longest wavelength) $\frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{658 \times 10^{-9}} = 3.0227 \times 10^{-19}\,\text{J}$ $E_3 - E_2 = 3.0227 \times 10^{-19}\,\text{J}$	 C1

	$E_3 = 3.0227 \times 10^{-19} + \left(-3.40 \times 1.60 \times 10^{-19}\right) = -2.42 \times 10^{-19}$ J	A1
7biii	Electric potential energy is negative due to negative work done by an external force in moving the electron from infinity to that point. This is a result of the attractive nature of the electric force between the negatively charged electron and the positively charged nucleus.)	B1
8ai	work function = hc/λ_0 where λ_0 is the threshold wavelength of gallium arsenide	